D 12425 (Pages : 4) Name.....

Reg. No.....

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THIRD SEMESTER B.Sc. DEGREE EXAMINATION, NOVEMBER 2016

(CUCBCSS–UG)

Complementary Course

MAT 3C 03—MATHEMATICS

Time : Three Hours Maximum : 80 Marks

Section A

Answer **all** *questions.*
Each question carries 1 *mark.*

1. Verify that $y = cx^3$ is a solution of $xy' = 3y$.
2. Solve $y' = -x/y$.
3. Test for exactness $2xydx + x^2dy = 0$.
4. Define rank of a matrix.
5. Find the characteristic roots of the matrix $A = \begin{bmatrix} -7 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 6 \end{bmatrix}$.
6. State the Cayley-Hamilton Theorem.
7. If $\vec{p}, [3, 2, 0]$ and $[-2, 4, 0]$ are in equilibrium, find $\vec{p}$.
8. Find the gradient of the scalar function $f(x, y, z) = xyz$.
9. Illustrate commutativity of vector addition with an example.
10. Define a simply connected domain.
11. State Green's theorem in the plane.
12. Give a parametric representation of the sphere.

(12 × 1 = 12 marks)

Turn over

Section B

Answer any **nine** *questions.*
Each question carries 2 marks.

13. Solve $\left(x^2+y\right)dx+\left(y^2+x\right)dy=0$.

14. Solve $y'-y=4$.

15. What is the characteristic equation of $\begin{bmatrix}1 & 2\\ 7 & 1\end{bmatrix}$?

16. Solve $dy/dx+y/x=x^3$.

17. Using Cayley-Hamilton theorem, find A^2 if $A=\begin{bmatrix}1 & 2\\ 1 & 1\end{bmatrix}$.

18. Find the rank of $A=\begin{bmatrix}1 & 2 & 3\\ 2 & 4 & 5\end{bmatrix}$.

19. Find a parametric representation of the straight line through (4, 2, 0) in the direction of [1, 1, 0].

20. Find a tangent vector and unit tangent vector for $\vec{r}(t)=\left[t,t^3,0\right]$.

21. Show that the form under the integral sign is exact :

$$\int e^{x-y+z^2}\left(dx-dy+2z\,dz\right).$$

22. Give the standard form of the Bernoulli equation with a suitable example.

23. Using Green's theorem, find the area enclosed by the ellipse :

$$x^2/a^2+y^2/b^2=1.$$

24. Find curl of $\left[y/z, x/z, -xy/z^2\right]$.

(9 × 2 = 18 marks)

Section C

Answer any **six** *questions.*
Each question carries 5 *marks.*

25. Find an integrating factor and solve :

$$\left(x^2-2x+2y^2\right)dx+2xy\,dy=0.$$

26. Find the Orthogonal Trajectories of $y=ce^{-x}$.

27. Reduce to normal form and find the rank of $A=\begin{bmatrix}1&1&1&1\\1&2&3&4\\4&3&2&1\end{bmatrix}$.

28. Solve : $x+2y+3z=0$.
$2x+y+3z=0$
$3x+2y+z=0.$

29. Verify Cayley-Hamilton theorem for $A=\begin{bmatrix}2&-1&1\\-1&2&-1\\1&-1&2\end{bmatrix}$.

30. Find the length of the vector $\vec{r}(t)=[t,\cosh t,0]$ from $t=0$ to $t=1$.

31. (i) Prove that curl (grad f) $=\vec{0}$ for any twice differentiable scalar function 'f'.

(ii) Prove that div (curl $\vec{v}$) $=0$ for any vector $\vec{v}$.

32. Use Gauss Divergence Theorem to evaluate :

$$\iint_S \vec{F}\circ\vec{n}\,dA\,;\vec{F}=\left[x^2,0,z^2\right]\ \text{S is the box}\ |x|\le 1,|y|\le 3,|z|\le 2.$$

33. Find the work done by a Force $\vec{F}=\left[y^2,-x^2,0\right]$ in moving an object along the straight line segment from (0, 0) to (1, 4).

(6 × 5 = 30 marks)

Section D

Answer any **two** *questions.*
Each question carries 10 *marks.*

34. Find the characteristic roots and any two characteristic vectors of :

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}.$$

35. (i) Find the tangential and normal acceleration of $\vec{r}(t) = \left[0, 0, 5t^2\right]$.

(ii) Find the directional derivative of $f(x,y) = x^2 + y^2$ at (1, 1) in the direction of [2, – 4].

(iii) Given $\vec{v} = \text{grad}\, f$, find '$f$' if $\vec{v} = \left[\frac{y}{z}, \frac{x}{z}, \frac{-xy}{z^2}\right]$.

36. State Stokes Theorem. Verify it for :

$\vec{F} = \left[z^2, 5x, 0\right]$, S is the square $0 \le x \le 1, 0 \le y \le 1, z = 1$.

(2 × 10 = 20 marks)